Aryaman Gupta

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Research Objective

My research aims to make robot foundation models (RFMs) safe and reliable by turning their failures/anomalies into a learning signal. I develop methods that discover interpretable failure modes from deployment data, monitor reasoning-action misalignments at runtime, and steer them toward safe, successful task execution at test time with minimal human supervision. Going forward, I aim to build safety-reasoning models that generalize across tasks and embodiments, enabling RFMs to monitor and correct their behavior, and explore post-training RFMs with reinforcement learning using dense failure signals.

Education

- **Stanford University** — Stanford, CA
  Ph.D. in Aeronautics and Astronautics; Advisor: Somil Bansal — *Jan 2025 – Present*
- **Stanford University** — Stanford, CA
  M.S. in Aeronautics and Astronautics; Advisor: Somil Bansal — *Jan 2025 – Dec 2026 (Exp.)*
- **Indian Institute of Technology (BHU)** — Varanasi, India
  B.Tech. in Electronics Engineering; Advisor: Om Jee Pandey — *Nov 2020 – May 2024*

Publications

[1] **Aryaman Gupta***, Yusuf Umut Ciftci*, Somil Bansal, "Unsupervised Discovery of Failure Taxonomies from Deployment Logs", *IROS'26*

[2] **Aryaman Gupta***, Kaustav Chakraborty*, Somil Bansal, "Detecting and Mitigating System-Level Anomalies of Vision-Based Controllers", *ICRA'24*

[3] A.M. Ali, **Aryaman Gupta**, H.A. Hashim, "Deep Reinforcement Learning for Sim-to-Real Policy Transfer of VTOL-UAVs Offshore Docking Operations", *Applied Soft Computing Journal*

[4] Neha Sharma, **Aryaman Gupta**, Sivala Deepak, Om Jee Pandey, "Node Fault Prediction Assisted Small-World IoT Networks Using ML Frameworks", *IEEE ANTS'24* [Best Paper Award]

Preprints

[1] **Aryaman Gupta**, Xiaoyue Wu, Rana Soltani Zarrin, Mark Van der Merwe, "GUARD: Anomaly-Informed Q-Steering for Generative Robot Policies in Dexterous Manipulation", *arXiv '26*

[2] Sathwik Karnik*, Joseph Lee*, **Aryaman Gupta**, Somil Bansal, "When Reasoning Helps Action: Monitoring and Steering Chain-of-Thought in Vision-Language-Action Policies", *arXiv '26*

[3] Kaustav Chakraborty, **Aryaman Gupta**, Somil Bansal, "Enhancing Safety and Robustness of Vision-Based Controllers via Reachability Analysis", *arXiv '24*

*Equal Contribution

Experience

- **Safe and Intelligent Autonomy Lab** — Stanford, CA
  Graduate Student | Prof. Somil Bansal — *Aug 2024 - Present*
  - **Research Area**: Policy Steering, Runtime Failure Monitoring, Failure-Guided Policy Improvement.
  - Designed test-time monitors that flag VLA reasoning-action misalignment and steer towards visually accurate chain-of-thought and actions, improving its performance, reliability, and human interpretability.
  - Developed an unsupervised multimodal-LLM pipeline that discovers interpretable failure taxonomies from deployment logs, enabling targeted policy improvement and runtime failure monitoring.
  - Designed Q-guided safety filtering for high-speed systems, such as F1-Tenth cars and drones operating under visuomotor policies, solely through perception inputs without privileged information.

- **Honda Research Institute - USA** San Jose, CA
  Research Internship *June 2026 - Sept 2026*
  - **Research Area**: Anomaly-informed test-time steering of generative policies for dexterous manipulation.
  - Proposed GUARD, which learns a Q-function from sparse task rewards and dense uncertainty scores, and uses its gradients to steer a frozen flow-matching policy toward successful, in-distribution actions.
  - Designed an offline-to-online Q-learning scheme that adapts the Q-function to the steered policy.
  - Achieved 96%/80% average success on four Adroit tasks under nominal/perturbed dynamics, outperforming the strongest baseline by 21/20 pp while keeping the lowest uncertainty score.
  - Demonstrated zero-shot sim-to-real transfer on in-hand screwdriver rotation and reorientation tasks.
- **University of Southern California** Los Angeles, CA
  Research Internship | Prof. Somil Bansal *May 2023 - Aug 2024*
  - **Goal:** Provide safety guarantees on vision-based controllers leveraging knowledge of their failures.
  - Mined failure modes of an aircraft taxiing system by computing Neural Reachable Tubes (NRTs).
  - Developed a runtime vision-based failure monitor and fallback mechanisms, reducing failures by 42%.
  - Fine-tuned policy on augmented failure dataset, reducing failures by 20% and prediction error by 10%.
  - Applied Conformal Prediction for probabilistic safety guarantees enabling deployment under uncertainty.
- **Carleton University** Ottawa, Canada
  Research Internship | Prof. Hashim Mohamed *Jan 2023 - Dec 2023*
  - **Goal:** Develop docking mechanism for VTOL-UAVs on offshore charging platforms using Deep-RL.
  - Built a custom environment for UAV landing with JONSWAP model-based hydrodynamic disturbances (waves) acting on the docking station, making it oscillate on the water surface.
  - Implemented DQN and PPO algorithms to compare performance among value and policy-based agents.
  - Agents successfully learnt to land precisely and safely through a carefully designed reward function.
- **Indian Institute of Science** Bengaluru, India
  Research Internship | Prof. Bharadwaj Amrutur *May 2022 - July 2022*
  - **Goal:** Develop centralized multi-agent exploration and vision-aided dynamic collision avoidance systems.
  - Implemented RRT-exploration algorithm and map merging on a multi-robot fleet for collective SLAM.
  - Performed real-time map updates, detecting obstacles with 3D Object Detection through infrastructure cameras and using Gradient Descent to optimize obstacles' 3D real-world position for avoidance.
  - Developed ROS packages that can be configured and implemented on custom hardware testbeds.
- **Changwon National University** Changwon, South Korea
  Research Internship | Prof. Oh-Seol Kwon *Mar 2022 - July 2022*
  - **Goal:** Develop a deep learning architecture for efficient object detection in low-resolution aerial images.
  - Combined Faster R-CNN, Edge Enhanced Network, and SRGAN architectures for the joint task.
  - Performed End-to-End Training of the entire pipeline, feeding detector's loss to SRGAN network.
  - Obtained testing accuracies of 95.5% on COWC and 83.2% on OGST datasets.

Selected Achievements

- Qualcomm Innovation Fellowship – among 16 selected teams out of 400+ applicants 2026
- Annenberg Scholarship, University of Southern California 2024
- Best Paper Award – IEEE Advanced Networks and Telecommunication Systems 2024
- Best Undergraduate Thesis – among 146 students in ECE IIT BHU 2024
- IUSSTF-Viterbi Scholarship – among top 15 students selected across India 2023
- DAAD-WISE Scholarship 2023
- Mitacs GRI Scholarship 2023
- Felicitation by IIT BHU for technical contributions, leadership, and mentorship to fellow students 2023
- 2nd Position – across India in All IITs Robotics Association Challenge 2021

Relevant Coursework

- **Graduate Courses**
  - CS-224N (Natural Language Processing)
  - AA-276 (Principles of Safety-Critical Autonomy)
  - AA-228V (Validation of Safety-Critical Systems)
  - ENGR-205 (Introduction to Control Design Techniques)
  - CS-224V (Agentic AI)
  - AA-228 (Decision Making Under Uncertainty)
  - AA-203 (Optimal and Learning-Based Control)
  - AA-242A (Classical Dynamics)
- **Undergraduate Courses**
  - MA-101 (Real Analysis)
  - MA-102 (Linear Algebra)
  - MA-202 (Probability and Statistics)
  - CSO-102 (Data Structures and Algorithms)
  - CSO-332 (Ubiquitous Computing)
  - CSO-458 (Soft Computing)

Skills and Interests

- **Languages and Libraries:** Python, C++, MATLAB, Julia, JAX, Hugging Face, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, OpenAI Gym, X-Plane, CARLA, MuJoCo, mjlab
- **Technologies and Tools:** ROS, ROS2, Flightmare, AutowareAI, Open3D, Linux, Git, Docker, LaTeX

Community Services and Involvements

- Reviewer – CoRL (2025, 2026), IROS (2026), RA-L (2025, 2026), T-RO (2025)
- Working Group Member – Stanford Center for AI Safety
- Technical Lead – RoboReG, a student-led robotics research group at IIT BHU

Student Mentorship

- Joyce Yang (Cornell BS CS), Tito Rosas (UCLA BS ME)